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A Deep Learning Based Approach for Predicting the Outcome of H-1B Visa Application

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Abstract. The H-1B is a visa that allows US employers to employ foreign workers in specialty occupations. The number of H-1B visa applicants is growing drastically. Due to a heavy increment in the number of applications, the lottery system has been introduced, since only a certain number of visas can be issued every year. But, before a Labor Condition Application (LCA) enters the lottery pool, it has to be approved by the US Department of Labor (DOL). The approval or denial of this visa application depends on a number of factors such as salary, work location, full time employment, etc. The purpose of this research is to predict the outcome of an applicant's H-1B visa application using Artificial Neural Networks and to compare the results with other Machine Learning approaches.

Keywords: H1B, Labor Condition Application, Deep Learning, Artificial Neural Networks

1 Introduction

The H-1B is a work visa granted by the United States (US) which allows an individual to temporarily work at an employer in a specialty occupation. According to the Immigration Act of 1990, a total of 65000 foreign nationals may be issued a visa each fiscal year. However, an additional 20,000 H-1Bs are available to foreign nationals holding a master's or a higher degree from US universities. An individual can stay in the US on an H-1B visa for a period of 3 years which is extendable up to six years. Although the final outcome of the visa status (approved or denied) is based on the lottery system, the LCA is scrutinized by the US DOL prior to entering the lottery pool. A number of factors such as the salary, job location, job designation, etc. are taken into account by the DOL to determine the approval or denial of the application. The application is forwarded to the lottery pool once it has been approved by the DOL. If denied, the applicant has to reapply before his/her current H-1B visa term expires.

At present, there is no such model which predicts the outcome of H1B visa

application. The applicant applies through a standard process and waits for the decision. If the visa is denied then the applicant has to reapply, which is in a way inefficient for the applicant. It is therefore essential to have an automated tool that could give insights to the applicant prior to applying.

Machine Learning (ML) based approaches are becoming popular for prediction problems in different domains including healthcare, banking, security, computer vision, natural language processing etc. Artificial Neural Networks (ANN), which is inspired by human brain, is a state of the art ML technique to find patterns in data and make predictions. Here, an automated system is proposed using ANN to predict the outcome of H-1B visa application by taking into account multiple factors such as salary, work location, job designation, etc. This will help the applicant to determine his/her chances of approval. This model has been compared with other ML algorithms to understand the overall efficiency.

2 Literature Review

A large number of changes are taking place while recruiting new employees due to the restrictions imposed by H-1B visa. For instance, Monica Boyd [1] has presented the changes in recruiting tracts because of the changes in H-1B policies. H-1B visa has played a key role in hiring international talent. Axel Dreher et al. [2] have showed how the US has managed to improve its post-secondary education by hiring top faculty from overseas. This shows how important an H-1B visa is to the employers.

The approval of H-1B visa application is decided on several parameters. In the current scenario, the salary parameter is extremely important due to some of the changes, which have been proposed by the administration, and as a result, it is crucial to understand the salary trends of the H-1B workers. Interesting salary versus case status trends have been presented by Renchi Liu et al. [3]. Renchi Liu et al. have used Decision Tree and K-means clustering to analyze the attributes of foreign workers such as salary, work location, etc. They have presented a model which shows a picture of different approval rates compared with different conditions based on the data. It is evident how salary changes from one state to another from the salary versus job location trends presented by Renchi Liu et al. These trends can be useful not just from the application point of view but also prior to finding a job. For example, a trend shows that people working in California are earning quite higher than any other state. This trend can be helpful for people who are looking for the highest paid job. **Interesting trends presented by Renchi Liu et al. [3] helped us in identifying the effect of parameters across the dataset.**

3 Problem

3.1 Problem Statement

Given the attributes of an applicant as input, determine the likelihood of H-1B visa approval.

3.2 Approach

The system has been designed in two phases

Phase 1: Data Pre-processing

Phase 2: Prediction of the H-1B visa application (Approved or denied)

4 Dataset and Data Pre-processing

The dataset was acquired through Kaggle [4]. The dataset has about 3 million entries. However, we truncated this dataset to a size of 1 million for testing our prototype. The name and description of each column is given below:

1. EMPLOYER NAME: Name of the Employer
2. YEAR: H-1B visa petition year
3. SOC NAME: Name associated with the SOC CODE, which is an occupational code, associated as the job has been granted temporary labor condition, this is required for Standard Occupational Classification (SOC) System
4. JOB TITLE: Title of the job
5. PREVAILING WAGE: Temporary labor condition is requested for the job and to set Prevailing Wages. The wage is scaled and listed in USD. The definitions on Prevailing wages are as follows: the average wage paid to similarly employed workers in the requested occupation in intended employment. It is an indication of employer's minimum requirements at the company for the specified job title
6. FULL TIME POSITION: Y = Full Time Position; N = Part Time Position
7. WORKSITE: Information of the applicant's city and state intended for work in USA
8. CASE STATUS: It states if an applicant has been granted a visa or not. (Certified implies that a visa has been granted)
9. LAT/LON: It is the latitude and longitude of the worksite of the employer

4.1 Chosen Attributes

The dataset we have used had data for the year 2016 only. Therefore, we decided to remove the YEAR attribute. We are trying to predict the likelihood of approval based on the geographical area the individual works in. The exact location (longitude and latitude) may adversely affect the generality of the algorithm. Therefore, we decided to remove the same and keep WORKSITE only in the dataset. The resulting dataset used for training and testing had the following attributes: EMPLOYER_NAME, SOC_NAME, JOB_TITLE, FULL_TIME_POSITION, PREVAILING_WAGE, WORKSITE and CASE_STATUS

4.2 Removing Outliers

The only numeric attribute in the dataset was PREVAILING WAGE. We used box-plot to detect the outliers and removed the same using Tukey's $1.5 \times \text{IQR}$ method [5].

4.3 Balancing Dataset

The original dataset had four possible values for the attribute CASE_STATUS, which were CERTIFIED, DENIED, WITHDRAWN and CERTIFIED-WITHDRAWN. We removed WITHDRAWN and CERTIFIED-WITHDRAWN rows from the dataset because it solely depends on the applicant. In addition, the original dataset was highly skewed. The number of certified rows was very less compared to that of denied. Therefore, we randomly selected rows from the dataset such that the resulting dataset had equal number of certified and denied rows to avoid bias.

4.4 One-Hot Encoding

Except PREVAILING WAGE, all other attributes were in textual format. To feed them to the neural networks, it was necessary to convert them into numeric format. FULL_TIME_POSITION column had only two possible values: Y and N. We replaced Y with 1 and N with 0. CASE_STATUS column also had only two possible values after removing withdrawn cases in the previous step. So we replaced CERTIFIED with 1 and DENIED with 0. The remaining four columns: EMPLOYER_NAME, SOC_NAME, JOB_TITLE, WORKSITE were textual. To get the numeric representation of these columns we used one hot encoding technique implemented in Keras library. The pictorial representation of the one-hot encoding is shown in Fig.1. Left side of the figure shows six columns. Four out of the six columns are then one-hot encoded individually. All the resultant one-hot encoded columns are merged to produce the final feature set of size 16978.

4.5 Normalization

After one-hot encoding, all attributes, except prevailing wage, had values 0 or 1. Therefore, we normalized prevailing wage using min-max normalization to scale its value in the range [0, 1]

$$y = \frac{x - \min\{X\}}{\max\{X\} - \min\{X\}}$$

x- An attribute value

X- An array of all possible values for attribute x y- Normalized value of attribute x

After preprocessing, we got a dataset having 26962 samples. This resulting dataset was used for training and testing.

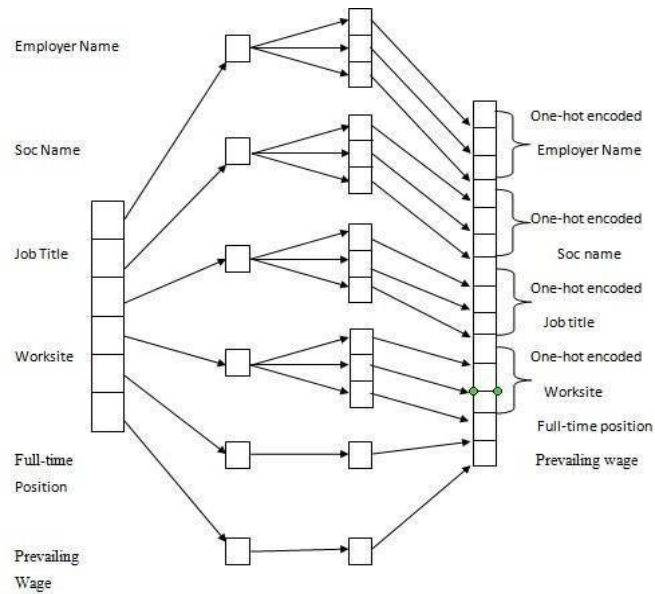


Fig. 1. The attributes chosen as input are on the left side of the figure. Each of these attributes is preprocessed separately. Some of them are one-hot encoded and others are kept as they were. Finally these are clubbed together to get the final attributes which will be fed to the model.

5 Experiments

Artificial Neural Networks

Artificial Neural Networks, which is inspired by human brain, is a state of the art technique to find patterns in data and make predictions. An ANN contains neurons connected to each other in different ways. A neuron basically represents an activation function, which transforms an input into output.

A linear activation function is easy to train but cannot be used for learning complex patterns. Non-linear functions, like sigmoid and hyperbolic tangent, can learn complex patterns in data but are sensitive to input only when it is in the middle range and they can saturate. This makes ANN containing these non-linear units difficult to train. Also, when the number of layers increases, it may result in vanishing gradient problem. Rectified linear function is a piecewise linear function that outputs the input as it is if it is positive and outputs zero if it is negative. The graph of the function is shown in Fig. 2. The rectified linear activation function and its derivative are easy to calculate and this makes ANN containing Rectified Linear Unit (ReLU) easy to train. The fact that it outputs 0 for negative values leads to sparsity, which is often desirable in representation learning. Because of the piecewise linearity, there is no gradient vanishing effect [6].

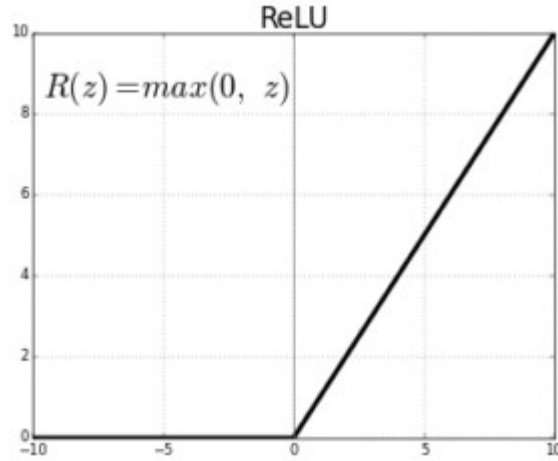


Fig. 2. Graph of rectified linear function for ReLU

We applied Multi-Layer Perceptrons (MLP) to our problem to predict the case status of a visa application. The ANN architecture we used is shown in Fig. 3. The input to

the neural network is the one-hot encoded data generated after preprocessing. The input layer is of size 16978. It has four hidden layers having neurons 512, 128, 64, 32 respectively of ReLU type and an output layer containing single neuron of sigmoid type. Visa status prediction being a binary classification problem, sigmoid was the suitable choice for the output layer. The output neuron gives a probability value of the case status. The output probability is then rounded off to get the final output i.e. if probability is 0.5 or <0.5 then it becomes 0, which means DENIED and if probability is >0.5 then it becomes 1, which means CERTIFIED. Since this is a binary classification problem, we used binary cross-entropy function. For training the ANN model, we experimented with Stochastic Gradient Descent and Adam [7] optimizers implemented in Keras library and chose Adam since it was converging faster.

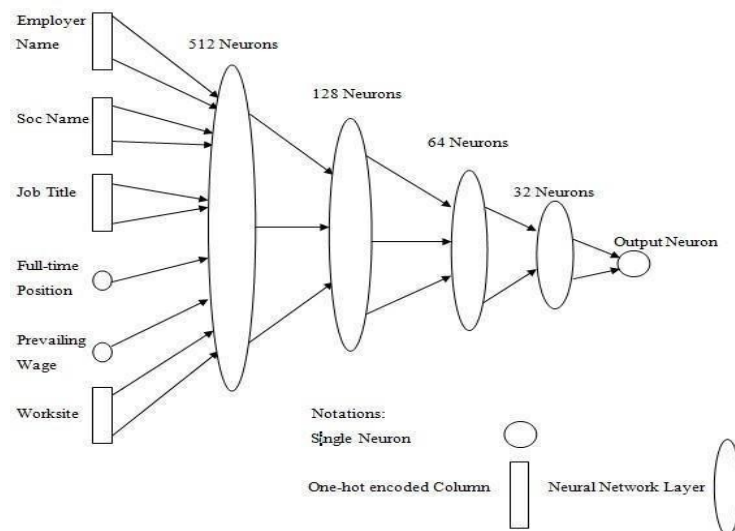


Fig. 3. Figure shows the architecture of the neural network used. The one-hot encoded input attributes are shown on the left side of the figure. Hidden layers of 512, 128, 64 and 32 neurons are used followed by an output layer consisting of single neuron which gives the prediction.

Other ML Algorithms:

We compared the outcome of ANN with other standard ML algorithms like Logistic Regression, Gaussian Naïve Bayes and Random Forest and found the prediction from ANN to be more accurate. The results are shown in Fig. 4 and Table 1.

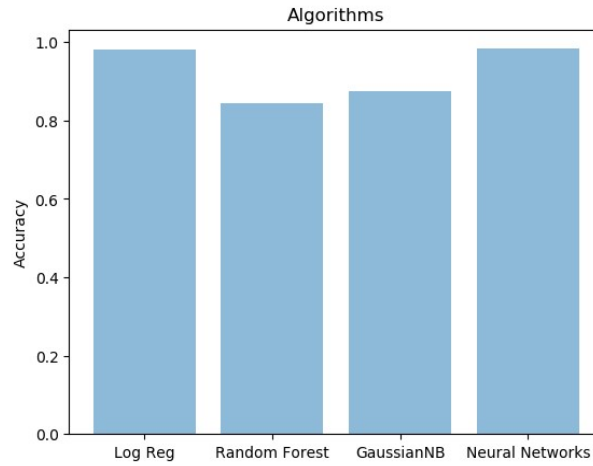


Fig. 4. Figure shows accuracy obtained using different Machine Learning algorithm

Table 1. Accuracy for H1B visa prediction task for different Machine Learning algorithms

Algorithm	Accuracy
Logistic Regression	96.27%
Random Forest Classifier	84.30%
Gaussian NB	87.50%
Neural Networks	98.3%

5.1 Training and Testing:

The entire dataset was split into training and testing in the ratio 80:20. For ANN, 20% of the training dataset was separated and used as validation data during training. The ANN model was trained on the training dataset for multiple epochs. After each epoch, the validation dataset was fed to the model to validate if the model is being trained properly or not. The training was stopped when the accuracy on the validation dataset stopped improving considerably. The trained ANN model was then used on the testing dataset for verification.

5.2 Results:

Fig. 4 shows accuracies obtained for different algorithms. The ROC plots for the logistic regression and ANN models are shown in Fig. 5. The area under the curve (AUC) value is 0.9629 for ANN and 0.9611 for logistic regression model. Fig. 6

shows the confusion matrix, precision-recall score and F1 score. It is clear from Fig. 6 that the false positive value is 16 for ANN and 68 for logistic regression model.

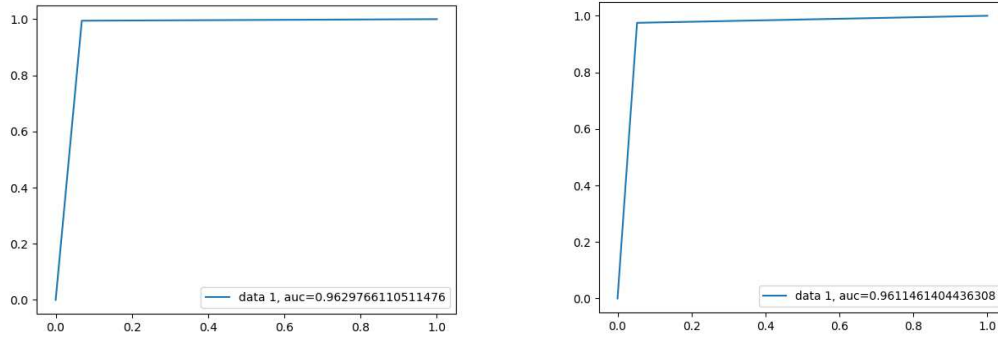


Fig. 5. ROC Curve for ANN (on left) and for Logistic Regression (on right)

The confusion matrix is			
	Predicted Positive	Predicted Negative	
Actual Positive	2720	200	
Actual Negative	16	2865	
Average precision-recall score: 0.93			
F1 Score for the model: 0.9636730575176591			

The confusion matrix is			
	Predicted Positive	Predicted Negative	
Actual Positive	2536	141	
Actual Negative	68	2648	
Average precision-recall score: 0.94			
F1 Score for the model: 0.9620345140781108			

Fig 6. Confusion Matrix and F1 Score for ANN (on left) and for Logistic Regression (on right)

6 Future Scope

In this paper we are just predicting whether the applicant's H-1B visa application will be approved or not. Also, we trained the model on a truncated dataset to test our prototype. However, in future we will be using the entire dataset and will be creating a recommendation system that will guide the applicant in the right direction. The recommendation system will generate recommendations if the prediction of the outcome of H-1B visa for an applicant is DENIED and will suggest improvisations in the applicant's profile to increase the chances of H-1B visa approval.

References

1. Monica Boyd: Recruiting high skill labour in north america: Policies, outcomes and futures. *International Migration*, 52(3):40–54, 2014
2. Axel Dreher, Panu Poutvaara: Student flows and migration: An empirical analysis. 2005
3. H-1B Visa Data Analysis and Prediction by using K-means Clustering and Decision Tree Algorithms. [Online] Available: <https://github.com/Jinglin-LI/H1B-Visa-Prediction-by-Machine-Learning-Algorithm/blob/master/H1B%20Prediction%20Research%20Report.pdf>
4. H-1B Visa Petitions 2011-2016 - Kaggle. [Online]. Available: <https://www.kaggle.com/nsharan/h-1b-visa/data>. [Accessed: 20-Oct-2017].
5. Seo, Songwon (2006) A Review and Comparison of Methods for Detecting Outliers in Univariate Data Sets. Master's Thesis, University of Pittsburgh (Unpublished).
6. Glorot, Xavier & Bordes, Antoine & Bengio, Y.. (2010). Deep Sparse Rectifier Neural Networks. *Journal of Machine Learning Research*. 15.
7. Diederik P. Kingma, Jimmy Ba Adam: A Method for Stochastic Optimization